

Understanding Is Not Observable

Shared Convention Makes Communication Possible
and Its Measurement Impossible

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Abstract

This paper began as an attempt to give brain-to-brain interface research what Shannon gave telecommunications: a governing quantity to aim at rather than a catalogue of devices to imitate. We define one — *coupling capacity*, a directed, interface-supremum generalisation of channel capacity over state trajectories — and prove that it saturates at the smaller of the two systems’ effective representational dimensionalities rather than at the channel rate. We then show that the quantity so defined cannot be measured, and that the obstruction is not a defect of our formulation but applies to every observational coupling measure in current use.

Four fields ask whether one system understands another — artificial intelligence of its models, neuroscience of interacting brains, comparative cognition of other species, philosophy of other minds — and all four answer the same way: observe what passes between the systems and infer from it. We argue that this cannot work, for two independent reasons.

First, we prove a non-identifiability result. Any two systems that can communicate must already share prior convention, and shared convention is a common cause of both what is transmitted and how the receiver behaves. There therefore exist dyads with *identical* observational distributions over signals, internal states and behaviour, one genuinely coupled and one not. No functional of the observational distribution — mutual information, transfer entropy, neural synchrony, representational similarity, behavioural agreement, or conversational performance — separates them. The very thing that makes communication possible is what makes it unmeasurable from observation. The construction is not a knife-edge coincidence: we exhibit an open set of linear–Gaussian dyads with the same property.

Second, and independently of confounding, we demonstrate with complete ground truth that coupling measured at the level of *internal states* does not imply coupling at the level of *behaviour*. In a trained two-agent system where the sender encodes its private state into the channel at $R^2 = 0.90$ and a linear decoder recovers that state from the receiver’s hidden layer to a reconstruction error of 0.0017, the receiver captures approximately 12% of what the information is worth and performs statistically at the goal-blind optimum. Removing the channel entirely and handing the receiver the sender’s state directly — infinite bandwidth, zero noise — changes nothing. We further find that randomising a signal is a materially more sensitive probe than ablating it, and that behavioural sensitivity and functional benefit come apart in turn, so interventional tests must be calibrated against the value of the information rather than against zero.

We then characterise what does suffice. Functional coupling is identified by adjustment when the shared convention is observed, by instrumental variation — for which, we note, ordinary channel noise qualifies, inverting the usual impulse to denoise — or by direct intervention. Two things notably do not suffice: temporal precedence, so that transfer entropy and Granger causality inherit the full force of the negative result despite being called directional; and observation of the receiver’s internal state, which fails the front-door criterion for structural reasons that no gain in recording resolution or interpretability precision can

repair. We conclude that functional coupling is an *interventional* quantity sitting one rung above where it is currently measured, and that this, rather than any deep obscurity in the phenomenon, is why debates about machine understanding do not resolve. We derive consequences for AI evaluation, mechanistic interpretability and hyperscanning, note that the framework retrodicts the decade-long stagnation of brain-to-brain interface bandwidth, and close with the open questions that would decide the account.

Keywords: functional coupling; causal identifiability; shared convention; coupling capacity; transfer entropy; emergent communication; hyperscanning; mechanistic interpretability; brain-to-brain interfaces; instrumental variables.

1 Introduction

A recurring question appears in several sciences at once, in slightly different dress each time.

Artificial intelligence asks whether a model understands an instruction, and answers with benchmarks, preference ratings and dialogue. Neuroscience asks whether two brains in interaction are coupled, and answers with inter-brain synchrony and directed information. Comparative cognition asks whether an animal grasps a signal. Developmental psychology asks it of infants. Philosophy asks it in general, calls it the problem of other minds, and has historically concluded that it is not empirically tractable.

Each field attacks the question with the same instrument: *watch what passes between the two systems, and infer understanding from the pattern*. Score the outputs. Measure the correlation between internal states. Decode one system’s content from the other’s activity.

This paper argues that the instrument cannot return the answer — not because measurement is noisy or the systems are complex, but because the quantity of interest is not a function of the data being collected. We give two independent arguments, one theoretical and one empirical, and one remedy.

How we arrived at this claim. We did not set out to prove a negative. The work began on the constructive side of the same question, in brain-to-brain interface research, which occupies today roughly the position telecommunications occupied before 1948: considerable engineering technique, no governing quantity that says what a link is for or how well it is doing. [Section 2](#) states the quantity we proposed to fill that gap — *coupling capacity*, a supremum of directed information over admissible interfaces — together with a bottleneck theorem showing it is capped by representational geometry rather than by bandwidth, and the simulation we built to test it. The programme did not survive its own controls, and it failed in a way that turned out to generalise far beyond the interface question that motivated it. We report it first because the rest of the paper is the diagnosis of that failure, and because the failure mode is, we will argue, shared by four literatures that have not noticed it.

The theoretical argument. [Section 4](#) is an identifiability result with an unusual structure. Ordinary confounding is a nuisance that careful design can sometimes remove. Here the confounder cannot be removed, because it is the precondition of the phenomenon. Two systems can only communicate if they already share a convention governing what signals mean. That shared convention is a common cause of what the sender emits and how the receiver behaves. Any observed coherence between them is therefore explicable *either* by the channel doing work *or* by both parties independently instantiating a convention they already held — and no

observational statistic separates these. The prerequisite for communication is the confounder for its measurement.

The empirical argument. Section 6 is independent and does not rely on confounding at all. In a fully observable artificial system with ground truth over every variable, we exhibit a receiver whose internal state provably contains the sender’s private information and whose behaviour provably extracts almost none of its value. State-level coupling is at ceiling; functional coupling is at floor. Since essentially all neural coupling measures and all decoding-based interpretability findings are state-level, this is a direct demonstration that such measures do not license behavioural conclusions.

The remedy. Section 8 follows from the diagnosis. If functional coupling is not identified by observation, it must be identified by intervention: ablate or randomise the channel and measure whether behaviour changes. What survives the removal of a signal is what the signal was doing. Everything else is fluency. We add one requirement that the interventional literature does not usually impose — that the effect be reported on the scale of the value of the information, not against a null — because our own data show sensitivity and benefit dissociating.

1.1 Contributions

1. A formal target for interface research — coupling capacity (Definition 1) and an effective-dimensionality bottleneck theorem (Theorem 2) — stated in full, together with an explicit account of why its empirical validation does not support it and why the quantity it bounds is not recoverable from observation (Section 2). The theorem stands as a statement about what a bandwidth-limited interface can drive; the measurement programme built on it does not.
2. A non-identifiability theorem for functional coupling under shared convention (Theorem 6), with the observation that the confounder is constitutive rather than incidental, and a linear-Gaussian family (Proposition 8) showing the construction occupies an open set of parameters rather than a measure-zero coincidence.
3. A matching positive result (Theorem 13) giving three sufficient conditions for identification, together with two sharp negatives: temporal-precedence measures such as transfer entropy do *not* escape the theorem (Proposition 16), and receiver-side measurement does not rescue it via the front-door criterion (Proposition 17).
4. A demonstrated instrumental-variable route — exogenous channel noise as an instrument (Remark 15) — validated on a controlled construction and then *falsified as a general recipe* on a trained system we did not design for it, with the mechanism of failure diagnosed rather than merely reported.
5. A ground-truth demonstration that state-level and behaviour-level coupling dissociate, including an infinite-bandwidth control in which the channel is deleted and the sender’s private state handed to the receiver directly (Proposition 19, Section 6).
6. A reporting statistic — the captured share of the value of information (Definition 21) — with an estimator, a confidence procedure and a checklist, motivated by our finding that a receiver can be measurably perturbed by a signal while deriving no benefit from it.

7. An account of why the gap arises: coupling is a jointly held convention, an attractor of shared history, which cannot be established unilaterally (Section 7); and a consequent prediction of hysteresis.
8. Consequences for AI evaluation, interpretability and hyperscanning (Section 9), and a retrodiction: the framework explains the decade-long flatness of brain-to-brain interface throughput against large hardware gains, and locates the error in the bandwidth argument that motivates such interfaces (Section 9.4).
9. An explicit register of open questions (Section 12), each stated so that a determinate answer would change the account.

1.2 Scope, and what we do not claim

We are not claiming to have solved semantics, and we do not use “understanding” in a phenomenal or philosophical sense. Throughout, *functional coupling* names a specific causal quantity: the extent to which one system’s signal changes what another system does. Our claim is that this modest, operational quantity — not consciousness, not meaning in any rich sense — is already unidentifiable by the methods four fields currently use, and that this is enough to invalidate a substantial body of inference.

Three further boundaries should be stated at the outset rather than left to Section 11. (i) Theorem 6 is a statement about what observational data cannot determine *absent further assumptions*; it is not a claim that coupling can never be estimated, and Theorem 13 says exactly when it can. (ii) Proposition 19 is an existence proof on one trained system; it establishes that the dissociation is possible and shows a mechanism by which it arises, not that it is prevalent. (iii) The interventional protocol of Section 8 has been applied to the artificial system reported here and not yet to human–AI or hyperscanning data; Section 12 states those as the principal outstanding obligations, with designs.

1.3 Roadmap

Section 2 states the coupling-capacity programme, its bottleneck theorem, and the controls that ended it; a reader interested only in the paper’s standing claims may begin at Section 3. Section 3 fixes notation and the causal object of interest. Section 4 proves non-identifiability. Section 5 characterises when identification is recoverable and when it is not. Section 6 reports the state–behaviour dissociation, Section 7 explains it, and Section 8 gives the measurement protocol. Section 9 draws consequences for four fields, Section 10 positions the argument against prior work, Section 11 states how it could be wrong, and Section 12 lists what would settle it. Proofs, experimental detail and a pre-registered protocol for the outstanding human–AI study appear in the appendices.

2 The target we set out to supply

Before 1948, telecommunications engineering possessed extensive practical technique but no rigorous definition of information or of a medium’s fundamental capacity. Shannon’s contribution [33] was not a device but a measurement theory: it specified the upper bound on error-free transmission over any noisy medium, and thereby gave a field a target to aim at rather than a catalogue of gadgets to imitate.

Brain-to-brain interface (BBI) and brain–computer interface research occupies an analogous position. Considerable ingenuity goes into tokenisation schemes, neural decoders and aligned

embedding spaces [13, 26, 29, 39], yet two questions have no agreed answer:

- (a) What quantity is being optimised across a link between two systems?
- (b) What bounds that quantity for two dynamic, non-stationary neural manifolds?

Absent answers, “success” is whichever downstream behavioural score a given paper reports, and results across hardware, species and modality cannot be placed on a common axis. This section states the quantity we proposed, the one theorem we proved about it, and — in [Sections 2.3](#) and [2.4](#) — exactly what our validation did and did not establish.

2.1 Coupling capacity

Let each system i be characterised by a state space $\mathcal{M}_i$, a manifold carrying an internal state trajectory $x_i(t)$, together with a self-predictive model P_i that predicts $x_i(t + \Delta)$ from $x_i(\leq t)$. We deliberately do not assume $\mathcal{M}_i$ is shared, aligned, or even of comparable dimension across systems — a departure from hyperalignment approaches, which presuppose a common representational geometry.

An *interface* from i to j is a possibly stochastic map $g_{i \rightarrow j} : x_i(\leq t) \mapsto s_{i \rightarrow j}(t)$ producing a signal that influences the subsequent evolution of x_j , subject to a bandwidth or structural budget — a discrete channel of vocabulary size K , a continuous channel of bandwidth B , a hardware-constrained stimulation protocol. Write $\mathcal{G}$ for the admissible set under a given budget. The directed coupling under g at lag Δ is the transfer entropy [22, 30]

$$\text{TE}_{i \rightarrow j}(\Delta; g) = I(x_i(t); x_j(t + \Delta) | x_j(\leq t)), \quad (1)$$

the reduction in uncertainty about j ’s future beyond what j ’s own past supplies. The explicit lag matters because biological systems exhibit genuine transmission delay and representational drift, which a lag-free formulation would misattribute to noise.

Definition 1 (Coupling capacity). The coupling capacity from system i to system j is the supremum of directed coupling over admissible interfaces,

$$C(i \rightarrow j) = \sup_{g \in \mathcal{G}} \text{TE}_{i \rightarrow j}(\Delta; g). \quad (2)$$

This is the dynamical analogue of Shannon capacity: where Shannon’s is a supremum over input distributions of mutual information under a power constraint, coupling capacity is a supremum over *interface designs* of directed information between coupled dynamical systems. Total coupling is $C_{\text{total}} = C(i \rightarrow j) + C(j \rightarrow i)$, with an asymmetry index

$$A = \frac{C(i \rightarrow j) - C(j \rightarrow i)}{C(i \rightarrow j) + C(j \rightarrow i)} \in [-1, 1], \quad (3)$$

predicted to track task-defined role — leading versus following, speaking versus listening.

2.2 The dimensional bottleneck

The programme’s substantive claim was that coupling capacity is bounded by representational geometry rather than by the wire.

Theorem 2 (Effective-dimensionality bottleneck). *Let systems i and j have effective representational dimensionalities $d_i = \dim_{\text{eff}}(\mathcal{M}_i)$ and $d_j = \dim_{\text{eff}}(\mathcal{M}_j)$, and let the interface be a channel*

of scalar bandwidth B bits per step. Then $C(i \rightarrow j; B)$ is non-decreasing in B and

$$C(i \rightarrow j; B) \leq \min(B, \bar{c} \cdot \min(d_i, d_j)), \quad (4)$$

where $\bar{c}$ is a per-direction capacity in bits per independently drivable dimension per step. Consequently, as $B \rightarrow \infty$ the capacity saturates at $C^* = \bar{c} \cdot \min(d_i, d_j)$, independently of further bandwidth.

Proof. Bandwidth bound. The path $x_i(t) \rightarrow s_{i \rightarrow j}(t) \rightarrow x_j(t + \Delta)$ is a Markov chain, since x_j is influenced by x_i only through the transmitted signal. By the data processing inequality,

$$I(x_i(t); x_j(t + \Delta) \mid x_j(\leq t)) \leq I(s_{i \rightarrow j}(t); x_j(t + \Delta) \mid x_j(\leq t)) \leq B,$$

the last step because the channel carrying $s_{i \rightarrow j}$ has Shannon capacity B .

Dimensional bound. The exogenous drive enters x_j 's update through an injection operator whose image lies in the tangent space of $\mathcal{M}_j$, of dimension d_j ; it can excite at most d_j independent directions. Symmetrically the source can supply at most d_i independent predictive components, so the number of independently couplable directions is $\min(d_i, d_j)$. Each is a scalar coupled channel whose capacity is bounded by a finite constant $\bar{c}$ set by that direction's dynamic range relative to the receiver's intrinsic process noise. Summing over directions gives $C \leq \bar{c} \cdot \min(d_i, d_j)$, with no dependence on B .

Monotonicity. Enlarging B enlarges $\mathcal{G}$, and a supremum over a larger set cannot decrease. Taking the minimum of the two bounds and letting $B \rightarrow \infty$ gives the stated saturation. $\square$

Theorem 2 is, we believe, correct, and we do not withdraw it: it is a statement about what a bandwidth-limited interface can drive into a receiver of finite effective dimension, and its proof uses nothing beyond the data processing inequality and a tangent-space count. Concavity of $C(\cdot; B)$ in B — the smooth approach to the ceiling — follows from a time-sharing argument in the cases we checked; a general proof remains open. What does not survive is the measurement programme built on top of the theorem, for reasons given in [Section 2.4](#).

2.3 Three predictions, and what the simulation actually showed

The theorem and definition yield three falsifiable predictions.

- P1 — Saturation law.** $C(i \rightarrow j; B)$ increases monotonically and concavely in B , saturating at $\bar{c} \cdot \min(d_i, d_j)$ — that is, at the smaller system's *effective* dimensionality rather than at the channel's rate. Falsifier: capacity that keeps scaling with bandwidth well past that ceiling.
- P2 — Self-prediction governs efficiency.** The capacity achieved per unit of bandwidth, C/B , increases jointly in the two systems' self-predictive accuracies R_i and R_j : better internal world models extract more coupling from an identical channel.
- P3 — Asymmetry tracks role.** The asymmetry index (3) correlates with an externally defined task role across domains.

We tested these in a seeded sandbox of two coupled stationary vector-autoregressive processes joined by a quantised B -bit channel, with the receiver's effective dimensionality, the coupling gain and the bit budget all set by hand, reporting mean $\pm$ s.d. over five seeds under both a predictive-gain estimator and the model-free Kraskov–Stögbauer–Grassberger k -nearest-neighbour estimator [17]. All three predictions were supported. Capacity saturated far below the raw channel limit,

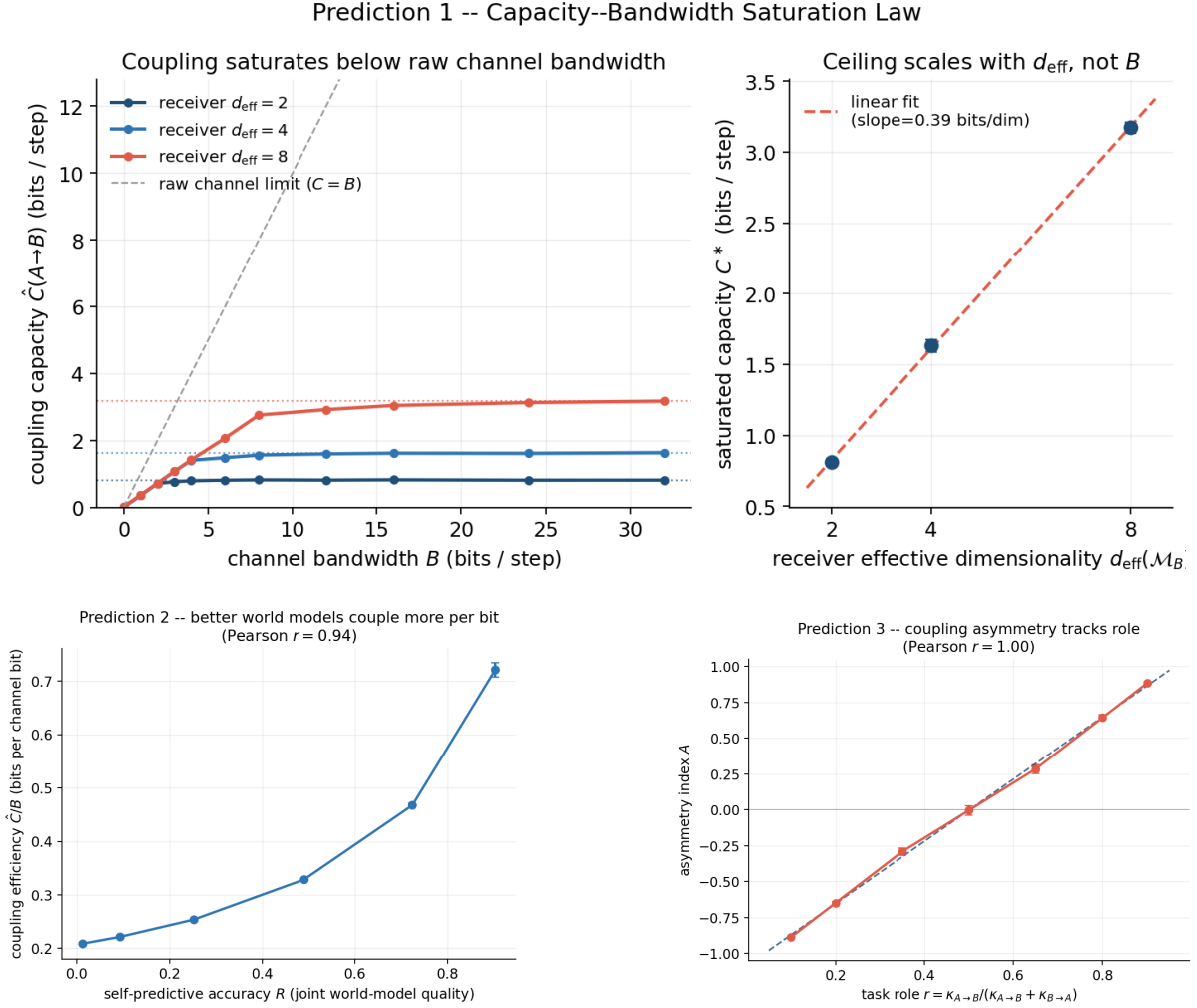


Figure 1: The superseded validation. Top: measured capacity against channel bandwidth for three receiver dimensionalities, saturating far below the raw channel limit (grey dashed), with ceilings linear in effective dimensionality at slope $\bar{c} \approx 0.39$. Bottom left: coupling efficiency C/B against joint self-predictive accuracy. Bottom right: asymmetry index against the imposed role parameter. Every panel behaves as Theorem 2 and P1–P3 require. Section 2.3 explains why this is nonetheless not evidence for the theory: the quantities being recovered were the ones written into the simulator.

with ceilings of 0.81, 1.63 and 3.17 bits/step at $d_{\text{eff}} = 2, 4, 8$, i.e. a slope of $\bar{c} \approx 0.39$ bits per dimension, and the ceiling tracked the hidden rank r rather than the ambient dimension when a rank- r manifold was embedded in a 16-dimensional state. Coupling efficiency C/B rose from 0.21 to 0.72 as self-predictive accuracy rose towards 0.9 (Pearson $r = 0.94$). The asymmetry index tracked the role parameter from -0.89 through 0 to $+0.88$ (Pearson $r = 1.00$). The two estimators, one parametric and one geometric, agreed to within 2% (Pearson $r > 0.99$).

What this established. Less than it appeared to, and we state the shortfall plainly because the rest of the paper is about what follows from it.

The simulation satisfies the theory’s assumptions by construction: it is linear, Gaussian and stationary, its manifolds have a dimensionality we chose, and its coupling has a gain we set. Under P3 in particular, a coupling parameter was imposed on the simulator and then recovered with $r = 1.00$; that is a demonstration that the estimator inverts the generative model, which is a property of the estimator, not evidence about the theory. The same reading applies with less

force to P1 and P2. A measurement theory does need to behave correctly where ground truth is known — this is a necessary condition — but we presented it at the time as confirmation, and it is not.

Two subsequent findings converted that overstatement into a structural problem.

The estimator produced coupling where there was none. Carried to a learned interface, the predictive-gain estimator reported a clean monotone bandwidth–coupling relationship at $r = 0.99$ — apparently P1 confirmed on a second, harder system. It was bias: the estimator is an in-sample regression whose bias grows with the ratio of channel dimension to sample size, and a pure-noise control returned 0.71 “bits” where the true value was zero. A block-shuffled surrogate correction removed essentially the entire effect, after which measured coupling sat at the noise floor across bandwidths 1–128. We return to this in [Remark 20](#), because it is the thesis of this paper occurring at the level of practice.

The receiver ignored the channel, and then ignored the answer handed to it. In the same learned-interface study the sender could be made to encode its private state at $R^2 = 0.90$ and the receiver’s hidden layer made to carry that state at reconstruction error 0.0017, with no behavioural consequence whatever; and a receiver given the sender’s state *directly*, with the channel deleted, behaved identically to one given nothing. [Section 6](#) reports this in full. Its immediate consequence here is that the bound in (4) was never the binding constraint on that system. [Theorem 2](#) is not contradicted — an unused channel is trivially consistent with any upper bound on what it could have driven — but the quantity the programme proposed to maximise turned out not to be the quantity that governs whether two systems accomplish anything together.

2.4 Why the programme could not be completed

The transfer entropy of (1) is a functional of the joint observational distribution of the two trajectories: in the vocabulary of [Definition 4](#) it is a Φ , and coupling capacity, being a supremum of such functionals over interfaces, is one too. [Theorem 6](#) therefore applies to it directly, and [Proposition 16](#) closes the escape that its directionality appears to offer. Two systems that can communicate at all must already share a convention; that convention is a common cause of what is sent and how the receiver acts; and no functional of the observational distribution — (1) included — separates a dyad in which the channel does the work from one in which it does not.

The programme was thus not defeated by an experimental disappointment that better tuning would repair. It was aimed at a quantity on the wrong rung of the causal hierarchy [27]. The remainder of this paper makes that claim precise, shows that it generalises well beyond interfaces, establishes a second and independent failure that does not involve confounding at all, and gives the interventional protocol that replaces the programme described above. [Section 9.4](#) returns to brain-to-brain interfaces with the diagnosis in hand, and finds that it retrodicts the field’s most conspicuous anomaly.

3 Formal setup

[Section 2](#) defined coupling over state trajectories, which is the natural object when the question is how much a channel can drive. The question of the rest of this paper is whether the channel drives anything at all, and for that a coarser object suffices. Consider a dyad consisting of a sender A and a receiver B exchanging signals over some channel.

Symbol	Meaning
Z_A	sender’s internal state (intent, neural state, private observation)
M	signal transmitted across the channel
Z_B	receiver’s internal state (activations, neural recording)
U	receiver’s behaviour (action, output, response)
C	convention-bearing latent shared by the dyad; generally unobserved
N	exogenous channel noise, candidate instrument (Remark 15)
Φ	an observational coupling measure: a functional of $P(Z_A, M, Z_B, U)$
$\mathcal{F}(\mathcal{D})$	functional coupling of the dyad (Definition 5)
$\text{do}(\cdot)$	Pearl’s intervention operator
ρ	captured share of the value of information (Definition 21)

Table 1: Notation used throughout.

Definition 3 (Dyad). A dyad is a tuple $\mathcal{D} = (Z_A, M, Z_B, U)$ of random variables, where Z_A is the sender’s internal state, M is the signal crossing the channel, Z_B is the receiver’s internal state and U is the receiver’s behaviour, together with a structural causal model specifying how each is generated [27].

We are deliberately permissive about what these denote. In hyperscanning, Z_A and Z_B are neural recordings and M is speech, gesture or gaze. In human–AI interaction, Z_A is the user’s intent, M the prompt, Z_B the model’s activations and U its output. In multi-agent learning they are agent states and messages. The argument is indifferent to substrate.

One further ingredient is deliberately not part of the tuple, precisely because it is generally unobserved: a convention-bearing latent C — the code, language or learned protocol the dyad already shares, without which M could not function as a signal at all. [Section 4](#) introduces it formally; it is flagged here because it is the reason the tuple above, exhaustive as it looks, does not determine $\mathcal{F}$.

Definition 4 (Observational measures). An *observational* coupling measure is any functional Φ of the joint distribution $P(Z_A, M, Z_B, U)$ obtained by passive observation of the dyad.

This class is broad and contains, to our knowledge, every measure currently in use: mutual information $I(Z_A; Z_B)$ and its lagged variants; transfer entropy $\text{TE}_{A \rightarrow B}$ [30] and Granger causality [9]; directed information [28]; the coupling capacity of [Definition 1](#); phase-locking and coherence measures in hyperscanning [23, 35]; representational similarity and cross-subject decoding; behavioural agreement scores; and performance on conversational benchmarks including the imitation game [40].

Definition 5 (Functional coupling). The functional coupling of a dyad is the causal dependence of the receiver’s behaviour on the signal:

$$\mathcal{F}(\mathcal{D}) > 0 \iff \exists m, m' : P(U \mid \text{do}(M=m)) \neq P(U \mid \text{do}(M=m')).$$

Where a magnitude is required we take

$$\mathcal{F}(\mathcal{D}) = \sup_{m, m'} D(P(U \mid \text{do}(M=m)) \parallel P(U \mid \text{do}(M=m')))$$

for a divergence D fixed in advance.

Functional coupling is thus an interventional quantity by construction: it lives on the second rung of the causal hierarchy [27], whereas every measure in [Definition 4](#) lives on the first. The

question this paper addresses is whether it can nonetheless be recovered from observational data — whether some clever Φ identifies $\mathcal{F}$. The answer is no, and the reason is specific to communication.

3.1 Assumptions

The results below rest on three assumptions, stated separately so that a reader who rejects one can see exactly which conclusions fall with it.

Assumption A1 (Convention precedes communication). For M to function as a signal, there exists a latent C , established prior to the exchange under study, such that the receiver's disposition to respond to M is a function of C . Equivalently, the edges $C \rightarrow M$ and $C \rightarrow U$ (via the receiver's dispositions) are present in the dyad's causal graph.

This is the substantive assumption, and it is the one the paper's negative results depend on. It is not an empirical hypothesis about particular dyads so much as a restatement of what it means for a signal to be a signal [18, 34]: a transmission in a code the receiver does not hold is not a degraded signal, it is not a signal.

Assumption A2 (C is not observed). The analyst does not record C , and no recorded variable is a sufficient adjustment set for it.

[Assumption A2](#) is what makes the problem hard rather than merely tedious; when it fails, [Theorem 13\(i\)](#) applies and identification is restored. [Remark 14](#) discusses how often it fails in practice, which is rarely.

Assumption A3 (Passivity). The analyst observes the dyad without setting M , and has no source of variation in M known to be independent of C .

[Assumption A3](#) defines the observational regime. [Theorem 13\(ii\)](#) and (iii) are exactly the two ways of violating it: find such a source of variation, or manufacture one.

4 Non-identifiability under shared convention

4.1 Why the confounder cannot be removed

Communication requires prior agreement. For a signal to carry information to a receiver, the receiver must already possess a convention — learned, evolved, trained or installed — that renders the signal actionable ([Assumption A1](#)).

This creates an unusual epistemic situation. The shared convention C is a common cause: it shapes what the sender emits ($C \rightarrow M$) and it shapes how the receiver behaves ($C \rightarrow U$), because the receiver's dispositions were formed under the same convention. Observed coherence between M and U is therefore ambiguous between two explanations that are ordinarily distinguished by design — and cannot be here, because eliminating C eliminates the possibility of communication along with the confound.

[Figure 2](#) makes the ambiguity concrete before the theorem states it formally. The two dyads constructed in the proof below differ only in whether M lies *on* the causal path from C to U or merely *beside* it. Both produce a message correlated with behaviour — in the left dyad because the message causes the behaviour, in the right dyad only because both are effects of the same shared convention. No inspection of (C, M, U) alone tells these apart; the difference appears only under intervention on M .

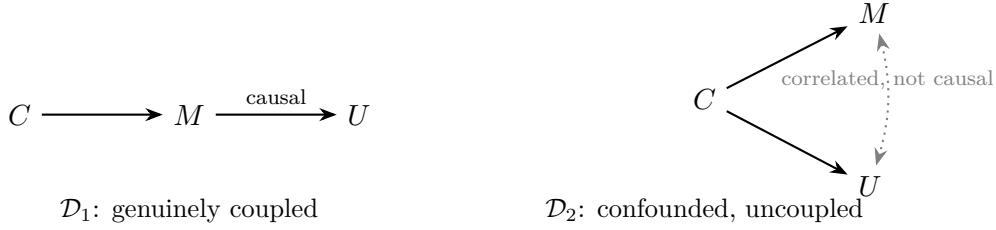


Figure 2: Both dyads induce the identical joint law over (C, M, U) used in the proof of [Theorem 6](#), so every observational functional Φ agrees on them. On the left, M mediates the only path from C to U : intervening on M changes U . On the right, U reads off C directly; M and U are statistically associated (dotted) purely through their common cause, and intervening on M leaves U unchanged. $\mathcal{F}$ tracks the solid causal arrow into U , not the association.

4.2 The result

Theorem 6 (Non-identifiability of functional coupling). *Under [Assumptions A1–A3](#) there exist dyads $\mathcal{D}_1$ and $\mathcal{D}_2$ such that*

$$P_1(Z_A, M, Z_B, U) = P_2(Z_A, M, Z_B, U) \quad \text{yet} \quad \mathcal{F}(\mathcal{D}_1) > 0 = \mathcal{F}(\mathcal{D}_2).$$

Consequently no observational measure Φ identifies $\mathcal{F}$.

Proof. Let $C \sim \text{Uniform}\{1, \dots, K\}$ with $K \geq 2$ denote a shared convention-bearing latent: the content the dyad has learned to coordinate on.

Dyad $\mathcal{D}_1$ (genuinely coupled). The sender’s state is $Z_A = C$; it emits $M = C$; the receiver’s state is $Z_B = M$ and its behaviour is $U = M$. Here B reads the channel and acts on what it reads.

Dyad $\mathcal{D}_2$ (confounded, uncoupled). The sender’s state is $Z_A = C$ and it emits $M = C$ exactly as before. The receiver, however, has independent access to C through shared prior context, and sets $Z_B = C$, $U = C$, ignoring M entirely.

Both dyads induce the same observational law: $Z_A = M = Z_B = U$ almost surely, uniform on $\{1, \dots, K\}$. Hence $P_1 = P_2$ and every functional of the observational distribution agrees on them. In particular $I(Z_A; Z_B) = I(M; U) = \log K$ in both — both dyads look maximally coupled by every available measure, including measures defined on behaviour.

Now intervene. In $\mathcal{D}_1$, setting $\text{do}(M = m')$ yields $U = m'$, so $P(U \mid \text{do}(M = m))$ varies with m and $\mathcal{F}(\mathcal{D}_1) > 0$. In $\mathcal{D}_2$, setting $\text{do}(M = m')$ leaves $U = C$ unchanged, so $P(U \mid \text{do}(M = m))$ is constant in m and $\mathcal{F}(\mathcal{D}_2) = 0$.

Since Φ depends on the dyad only through P , and $P_1 = P_2$ while $\mathcal{F}_1 \neq \mathcal{F}_2$, no Φ can equal $\mathcal{F}$. $\square$

Remark 7. The construction is elementary; that is the point. The mathematical content is standard confounding. The substantive claim is about *where* it applies: not to a badly designed study that could be fixed, but to every dyad capable of communicating at all, because shared convention is a precondition rather than a nuisance. One cannot randomise away the common cause without destroying the phenomenon.

4.3 The construction is not a knife-edge

A natural first objection is that [Theorem 6](#) relies on a deterministic, noiseless, exactly-matched pair, and that generic dyads — noisy, continuous, not fine-tuned — would be separated by observation. They are not. The following exhibits the same phenomenon on an open set

of parameters in the standard linear–Gaussian family, which is also the family in which the instrumental-variable remedy of [Remark 15](#) is verified.

Proposition 8 (Generic observational equivalence). *Let $C \sim \mathcal{N}(0, \sigma_C^2)$ and $N \sim \mathcal{N}(0, \sigma_N^2)$ be independent with $\sigma_C^2, \sigma_N^2 > 0$, and let $M = C + N$. Define*

$$\mathcal{D}_1 : U = \kappa M + \varepsilon, \quad \mathcal{D}_2 : U = \kappa' C + \varepsilon', \quad \kappa' = \kappa \frac{\sigma_C^2 + \sigma_N^2}{\sigma_C^2},$$

with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and $\varepsilon' \sim \mathcal{N}(0, \sigma_{\varepsilon'}^2)$ independent of everything else. If

$$\sigma_\varepsilon^2 > \kappa^2 \frac{(\sigma_C^2 + \sigma_N^2) \sigma_N^2}{\sigma_C^2}, \quad (5)$$

then setting $\sigma_{\varepsilon'}^2 = \sigma_\varepsilon^2 - \kappa^2(\sigma_C^2 + \sigma_N^2)\sigma_N^2/\sigma_C^2$ makes the observable joint laws coincide, $P_1(M, U) = P_2(M, U)$, while $\mathcal{F}(\mathcal{D}_1) > 0 = \mathcal{F}(\mathcal{D}_2)$. Since (5) is a strict inequality, the equivalence holds on an open subset of $(\kappa, \sigma_C^2, \sigma_N^2, \sigma_\varepsilon^2)$ -space.

[Section A](#) gives the calculation. Two consequences are worth stating.

First, the observationally identical pair is *stable under perturbation*: the equivalence does not require exact tuning of κ , only that (5) hold, so it cannot be dismissed as a measure-zero artefact. Second, the ordinary least-squares regression of U on M returns κ in *both* dyads, even though the true interventional effect is κ in one and 0 in the other — which is precisely what the simulation reported in [Remark 15](#) finds (0.800 ± 0.005 versus 0.799 ± 0.019 at $\kappa = 0.8$).

Remark 9 (Why C must be unobserved). [Proposition 8](#) matches the law of the observable pair (M, U) and not of (C, M, U) ; indeed $\text{Cov}(C, U)$ differs across the two dyads. This is not a weakness of the construction but a consistency check on it: if C were observed, [Theorem 13\(i\)](#) would identify $\mathcal{F}$ by back-door adjustment, and no observationally equivalent pair could exist. The equivalence and the identification result are two sides of [Assumption A2](#). When internal states Z_A, Z_B are recorded but C is not — the usual case in neuroimaging and interpretability — the burden is carried by [Proposition 17](#) rather than by this proposition.

4.4 Corollaries

Remark 10 (The case that matters most). $\mathcal{D}_2$ is not a contrived pathology. It is a compact description of a large language model in conversation with a person. The model was trained on the recorded output of human convention; the user is a member of the culture that produced it. Both parties therefore hold C independently, and their exchanges cohere *whether or not* the user’s particular signal is doing causal work. Fluency is exactly what $\mathcal{D}_2$ predicts. This is why conversational impressions of understanding are not merely unreliable but uninformative in principle.

Corollary 11 (Invalidity of conversational evaluation). *Any evaluation procedure that scores a system solely on the distribution of exchanged signals and responses — the imitation game and its descendants, dialogue benchmarks, human preference ratings — is a functional of P and therefore fails to identify $\mathcal{F}$. Such procedures cannot distinguish $\mathcal{D}_1$ from $\mathcal{D}_2$ at any sample size.*

Corollary 12 (Optimisation is indifferent). *Training that maximises an observational score Φ applies pressure that is by construction identical on $\mathcal{D}_1$ and $\mathcal{D}_2$. Feedback schemes that rate outputs therefore cannot select for functional coupling except through whatever correlation happens to hold in the training distribution. This is not a claim that current systems lack functional coupling; it is the claim that the training signal does not distinguish, so any such coupling is incidental to the objective rather than selected by it.*

5 When identification is possible

[Theorem 6](#) is a negative result about observational data absent further assumptions. It is not the claim that functional coupling can never be estimated. This section characterises what suffices — and, more usefully for practice, what does not. The negative results in [Section 5.3](#) are the ones with consequences for existing literature.

5.1 Sufficient conditions

Theorem 13 (Identification). *Functional coupling $\mathcal{F}(\mathcal{D})$ is identified if any of the following holds. Informally: you can see the confounder and adjust for it; you can find something that jitters the message for reasons unrelated to the convention and use it as leverage; or you can set the message yourself.*

- (i) **Adjustment.** *Assumption A2 fails: the convention-bearing latent C is observed and satisfies the back-door criterion relative to (M, U) . Then*

$$P(U \mid \text{do}(M=m)) = \sum_c P(U \mid M=m, C=c) P(c).$$

- (ii) **Instrument.** *Assumption A3 fails benignly: there exists N with $N \rightarrow M$, $N \perp\!\!\!\perp C$, and N influencing U only through M . Under standard auxiliary assumptions — linearity, or monotonicity in the discrete case [\[1\]](#) — $\mathcal{F}$ is point-identified; absent them it is partially identified, with informative bounds [\[2\]](#).*

- (iii) **Intervention.** *Assumption A3 fails by design: the experimenter can set M directly. Then $\mathcal{F}$ is identified by [Definition 5](#).*

Proof. (i) is back-door adjustment and (ii) instrumental-variable identification, both standard [\[1, 27\]](#); (iii) is immediate from the definition. $\square$

Remark 14 (On (i): rarely available). Adjustment is the condition one would most like to have and least often does. Convention is precisely what is implicit, distributed and unarticulated between two parties; a researcher able to write down C in full has already solved a harder problem than the one at hand. Proxies exist — measured familiarity or shared history for human dyads, overlap between a user’s idiom and a model’s training distribution for human–AI pairs — but adjustment on a proxy yields partial control, not identification, and the residual confounding is of unknown sign.

5.2 Channel noise as an instrument

Remark 15 (An underexploited inversion). The instrumental route is the practically interesting one, because instruments in this setting are often already present and are currently discarded. *Exogenous channel noise is an instrument.* Transmission dropout, trial-to-trial jitter, latency variability, hardware artefact — any perturbation of the transmitted signal that is independent of the dyad’s shared convention satisfies the requirements of [Theorem 13\(ii\)](#). This inverts the usual methodological posture: noise in the channel, ordinarily treated as a nuisance to be minimised or regressed away, is exactly what renders functional coupling estimable without intervening. Studies that carefully denoise their channel may be discarding their only source of identification.

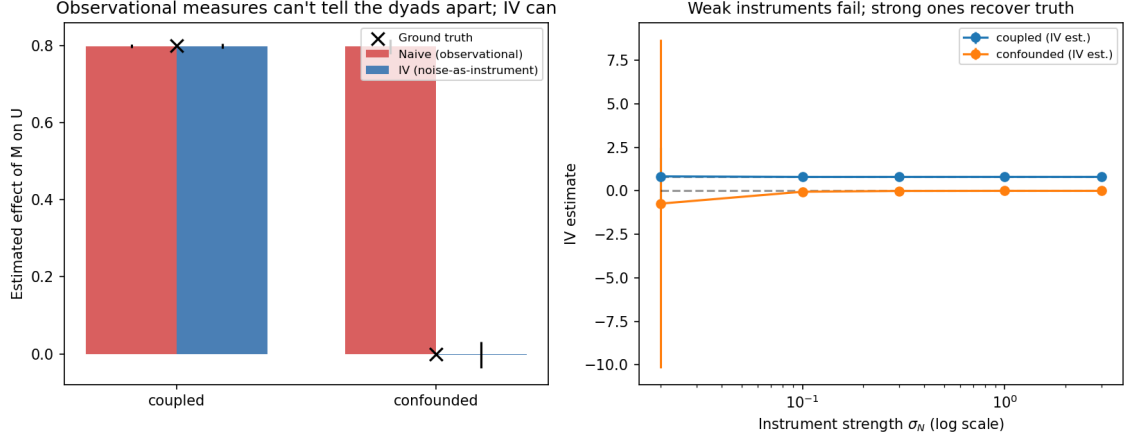


Figure 3: Left: the naive observational regression cannot separate the coupled dyad from the confounded one — both return the same estimate, and only the coupled dyad’s matches ground truth, by coincidence. The instrumental estimate recovers each dyad’s true value, including the confounded dyad’s true zero. Right: the instrument degrades gracefully as its strength σ_N falls, motivating the first-stage diagnostic rather than an assumed-valid instrument.

Verification on a controlled construction. We instantiate exactly the linear–Gaussian pair of [Proposition 8](#) — $M = C + N$ with N exogenous and independent of C , and U depending causally on M (coupled) or on C directly with M inert (confounded), calibrated so that $\text{Cov}(M, U)$ is identical across the two. The naive observational regression of U on M is statistically indistinguishable between them (0.800 ± 0.005 versus 0.799 ± 0.019 across 200 seeds, $n = 2000$ trials each), exactly as [Proposition 8](#) predicts. The instrumental estimate $\hat{\mathcal{F}} = \text{Cov}(U, N) / \text{Cov}(M, N)$ recovers 0.800 ± 0.007 for the coupled dyad and -0.003 ± 0.034 for the confounded one, matching each dyad’s true interventional effect. The observational gap between the two dyads is 0.0011; the instrumented gap is 0.8022, against a ground truth of 0.8.

This holds only once the instrument is strong enough. Sweeping the instrument’s strength $\sigma_N \in \{0.02, 0.1, 0.3, 1.0, 3.0\}$ and recording the first-stage F -statistic alongside each estimate: at $\sigma_N = 0.02$ ($F = 1.6$) the IV estimate is worthless, with a standard deviation larger than the effect itself; at $\sigma_N = 0.1$ ($F = 20.6$) it begins to work; by $\sigma_N \geq 0.3$ ($F > 180$) it is essentially exact. The conventional $F \gtrsim 10$ threshold [\[37\]](#) separates the two regimes cleanly here. The diagnostic must therefore be *reported* alongside the estimate, never assumed.

Failure on a system we did not build for it. Applying the identical method to the trained system of [Section 6](#) — using the discrete channel’s own per-step transmission noise as the instrument, over $150 \text{ goals} \times 20 \text{ noise redraws each}$, 3000 full episode rollouts — the instrument fails outright: first-stage $F = 0.2$, and the resulting IV estimate’s 95% confidence interval spans $[-715, 886]$, useless in either direction.

The failure is itself informative, and we can say why rather than merely that. A variance decomposition shows that across repeated noise draws for an identical goal the emitted message barely moves: within-goal variance is 0.3% of total message variance, while across-goal variance is 99.7%. The auxiliary loss that makes this channel informative ($R^2 \approx 0.90$; [Section 6](#)) trains the sender’s logits toward confident extremes, far from the threshold that the channel’s own noise perturbs — so the same optimisation that produces a usable code simultaneously starves that code’s transmission noise of instrumental relevance.

This is a genuine, checkable boundary condition on [Remark 15](#), not a defect in the controlled demonstration above. It generalises to a warning: well-trained discrete communication channels

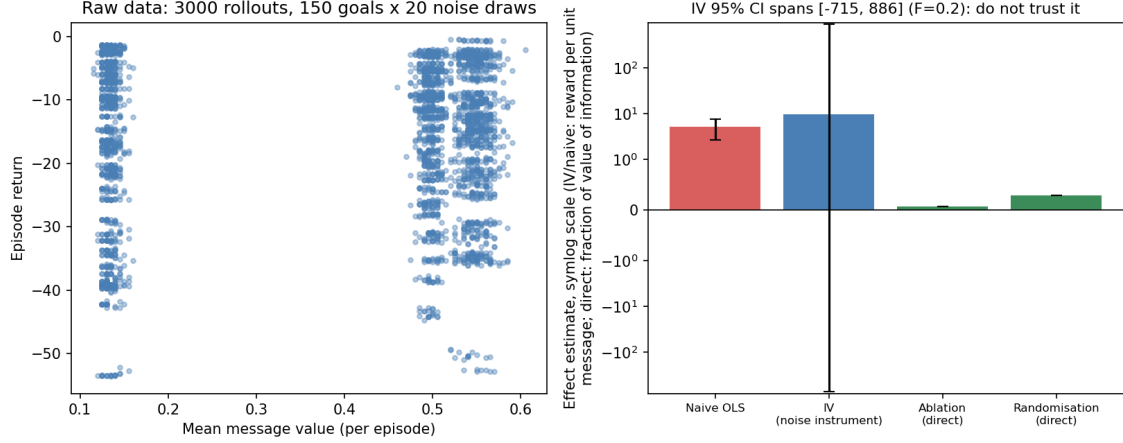


Figure 4: Left: raw rollouts on the real trained system. Message value clusters by goal rather than varying smoothly with noise — visible evidence of the low within-goal variance the text describes. Right: naive OLS and the IV estimate are both far from the ablation- and randomisation-based direct estimates of the value of information, and the IV estimate’s error bar is enormous. The first-stage diagnostic ($F = 0.2$) correctly flags the estimate as unusable rather than letting a point estimate mislead.

are a plausible common source of weak instruments, because optimisation pushes them towards low-entropy, saturated codes, and that is exactly the regime in which transmission noise stops being informative about the transmitted symbol — independently of whether functional coupling is present. Biological channels (speech, EEG artefact, stimulation jitter) are not optimised for threshold reliability in this way, so the specific failure mode may be milder there; but that is a claim to check, not to assume (Open Question 2).

5.3 What does not suffice

Proposition 16 (Temporal precedence is insufficient). *Measures defined by temporal ordering do not identify $\mathcal{F}$. Because C precedes both M and U , conditioning on the past of U does not block the back-door path $M \leftarrow Z_A \leftarrow C \rightarrow U$. Transfer entropy, Granger causality and directed information therefore remain functionals of the observational distribution and fall under Theorem 6 — as does any supremum of them over interfaces, including the coupling capacity of Definition 1.*

This deserves emphasis because such measures are routinely described as “directed” or “causal,” and the governing intuition is that respecting temporal order supplies causal content. It does so only in the absence of a common cause preceding both variables. In dyadic communication such a common cause is not a hazard to be checked for but a structural guarantee: without prior shared convention there is no communication to measure. The field’s principal workhorse is thus subject to the theorem in full.

Proposition 17 (Receiver-side measurement is insufficient). *Observing the receiver’s internal state does not restore identification via the front-door criterion — Pearl’s route for identifying a causal effect through a mediator when the direct treatment–outcome relationship is itself confounded, which requires no back-door confounding between treatment and mediator and none between mediator and outcome except what the treatment already blocks. Consider Z_B as candidate mediator on $M \rightarrow Z_B \rightarrow U$. Front-door identification requires (a) no unblocked back-door path from M to Z_B , and (b) that every back-door path from Z_B to U be blocked by M . If $C \rightarrow Z_B$*

and $C \rightarrow U$, then $M \leftarrow Z_A \leftarrow C \rightarrow Z_B$ violates (a), and $Z_B \leftarrow C \rightarrow U$ — which conditioning on M does not block — violates (b).

Both edges are present whenever the receiver’s internal representations and its behavioural dispositions were shaped by the same prior convention, which is to say whenever the receiver is capable of communicating at all.

Corollary 18. *Neither higher-resolution neural recording nor finer activation-level interpretability restores identification. Proposition 17 concerns causal structure, not measurement precision, and is therefore invariant to improvements in either.*

Corollary 18 combines with the empirical result of the next section to give receiver-side measurement a double failure: it targets the wrong node, and it cannot be adjusted into targeting the right one.

5.4 When nothing suffices: partial identification

A study may satisfy none of Theorem 13’s three conditions and still be worth analysing. Point identification is not the only useful outcome. Two standard tools apply directly and, to our knowledge, are not currently used in this literature.

If U is bounded — as it is whenever behaviour is scored on a finite scale, a fixed reward range, or a rubric — then even with no assumptions beyond the observed law, $\mathcal{F}$ obeys nonparametric bounds of the Manski type [20]: the interventional contrast is pinned within an interval whose width is set by the range of U and the support of M . The interval always contains zero under Assumption A2, which is Theorem 6 restated, but it is not the whole real line, and its width is a reportable measure of how much the design leaves undetermined.

If a candidate instrument is available but its exclusion restriction is only partially credible — the common case for channel noise in a corpus not designed around it — the Balke–Pearl bounds [2] give the tightest interval consistent with the observed law and a discrete instrument, without the monotonicity assumption that point identification requires. We recommend reporting these bounds rather than an unbounded point estimate whenever the first-stage diagnostic is marginal, since that is exactly the regime in which our own Stage-2 attempt above produced a confident-looking number spanning three orders of magnitude.

6 The state–behaviour gap

Theorem 6 concerns confounding. We now show a second, logically independent failure: even with no confounding and complete ground truth, coupling measured at the level of internal states does not imply functional coupling.

Proposition 19 (State–behaviour gap). *There exist dyads in which $I(M; Z_B)$ is near its maximum while $\mathcal{F}$ is negligible relative to the value of the transmitted information.*

We establish this constructively, in a system where every variable is observable and the causal structure is known by construction.

6.1 System

Two neural-network agents were trained jointly in a referential navigation task (PettingZoo simple_speaker_listener, 38). A *speaker* observes a private goal — which of three landmarks the partner must reach — and cannot move. A *listener* observes its own position and the landmarks

Quantity	Measured	Interpretation
Encoding fidelity (speaker $\rightarrow$ message)	$R^2 = 0.90$	message carries the goal
Representational uptake (message $\rightarrow$ receiver state)	recon. error 0.0017	goal present in receiver
Message sensitivity of policy (KL, real vs. shuffled)	0.021 nats	
Own-state sensitivity of policy (KL, reference scale)	0.686 nats	$\sim 33\times$ larger
Functional coupling — <i>ablation</i> (message zeroed)	$z = +0.50$	no detectable effect
Functional coupling — <i>randomisation</i> (message mispaired)	$z = +1.82$	small, marginal effect
Captured share of the value of information, $\hat{\rho}$	≈ 0.12	at the goal-blind optimum

Standardised effects z are computed against the intact condition. $\hat{\rho}$ is defined in [Definition 21](#) and computed from the anchors in [Table 3](#); it is not distinguishable from zero at the evaluation sample size.

Table 2: State-level coupling at ceiling; functional coupling small at best, and yielding no task benefit.

but not the goal, and must navigate. The environment’s fixed vocabulary was bypassed in favour of a learned discrete channel of tunable width, using a straight-through Gumbel–Sigmoid relaxation so that gradients flow from the listener’s loss into the speaker’s message [6, 12]. Agents were trained with policy gradients and an auxiliary reconstruction objective that pressures the message to carry the goal. [Section B](#) gives the full configuration.

Three quantities were measured after training, on frozen policies:

- **Encoding fidelity** — R^2 of a regression predicting the emitted message from the speaker’s private goal. This is a *positive signalling* measure in the sense of Lowe et al. [19].
- **Representational uptake** — reconstruction error of a decoder reading the goal from the listener’s hidden layer, i.e. whether the goal is present and linearly available in the receiver’s internal state. This is the artificial analogue of cross-subject decoding in neuroimaging and of feature discovery in interpretability.
- **Functional coupling** — the behavioural consequence of the message, estimated by intervention, in the two forms of [Section 8](#). *Ablation* replaces the message with zeros. *Randomisation* replaces it with a real, in-distribution message the speaker genuinely emitted on a different episode, preserving the message marginal exactly while destroying the message–state pairing. Both are direct estimates of [Definition 5](#); we report standardised effects z against the intact condition.

6.2 Result

The message demonstrably carries the goal. The goal is demonstrably present in the receiver’s internal representation, recoverable by a linear readout at negligible error. Under ablation, behaviour is statistically indistinguishable from behaviour with the channel zeroed.

Randomisation, however, is not null: destroying the message–state pairing while preserving the message marginal costs 2.32 reward units ($z = +1.82$), against 0.58 units for outright ablation ($z = +0.50$). Two conclusions follow, and the second matters more than the first.

Randomisation is the more sensitive instrument. It detects a residual behavioural dependence on message content that ablation misses — roughly a threefold larger standardised

Policy	Return	$\hat{\rho}$
Analytic expert (uses the goal optimally)	-8.78 ± 7.76	1.00 (anchor)
Goal-blind heuristic (hedges to landmark centroid)	-16.88 ± 9.84	0.00 (anchor)
Random	-41.94 ± 34.84	—
Trained agent, learned channel	-15.87	0.12
Trained agent, direct goal access (no channel)	-15.96 ± 9.70	0.11

$\hat{\rho}$ is the captured share of the value of information (Definition 21), computed against the expert and goal-blind anchors. Neither trained condition is distinguishable from the goal-blind anchor.

Table 3: An agent handed the goal for free converges to the goal-blind optimum.

effect on the same trained system. We had predicted this ordering on theoretical grounds (Section 8); it is here confirmed empirically. A zeroed channel presents a constant, easily accommodated input, whereas a mispaired but in-distribution message actively misleads any receiver retaining sensitivity to content. Practitioners adopting the protocol of Section 8 should prefer randomisation, and should not read a null ablation result as establishing zero functional coupling.

But the residual sensitivity does no work. The relevant scale is the value of the information itself. In this environment an analytic policy that uses the goal scores -8.78 and the best goal-blind policy scores -16.88 (Table 3), so knowing the goal is worth 8.10 reward units. The trained receiver scores -15.87 : statistically at the goal-blind optimum, capturing $\hat{\rho} \approx 0.12$ of the available value, and that residual is itself marginal. The receiver’s behaviour is faintly perturbable by the message while extracting essentially none of its value.

This is a sharper statement of the dissociation than a flat null would have been. *Behavioural sensitivity and functional benefit are themselves distinct.* A receiver can be measurably influenced by a signal and still derive no advantage from it — which means that even an interventional measure must be calibrated against what the information is worth, not merely tested against zero. We accordingly report $\hat{\rho}$ as the primary quantity (Definition 21) and recommend it over a bare significance test.

6.3 The infinite-bandwidth control

A natural objection is that some residual channel imperfection is responsible. We removed the channel.

In a separate condition the listener was given the speaker’s private goal *directly*, concatenated to its own observation — no encoder, no bottleneck, no discretisation, no noise. This is the limiting case of a perfect interface, and no physical channel can exceed it.

The agent with free, perfect, noiseless access to the information converged to -15.96 : the score of the goal-blind heuristic, indistinguishable from agents that received nothing. An analytic policy that uses the goal achieves roughly twice the return, so the information was both available and valuable. It was not used, and further training did not change this.

This is the strongest form of the dissociation. The failure is not attributable to bandwidth, noise, encoding, alignment or capacity, because in this condition none of those exist.

Remark 20 (A measurement artefact encountered en route). During this work an earlier analysis reported a clean monotone relationship between channel bandwidth and measured transfer entropy ($r = 0.99$), apparently confirming a capacity law. It was an artefact: the estimator is an in-sample regression whose bias grows with the ratio of channel dimension to sample size, and a pure-noise control returned 0.71 “bits” of coupling where the true value was zero. Applying a

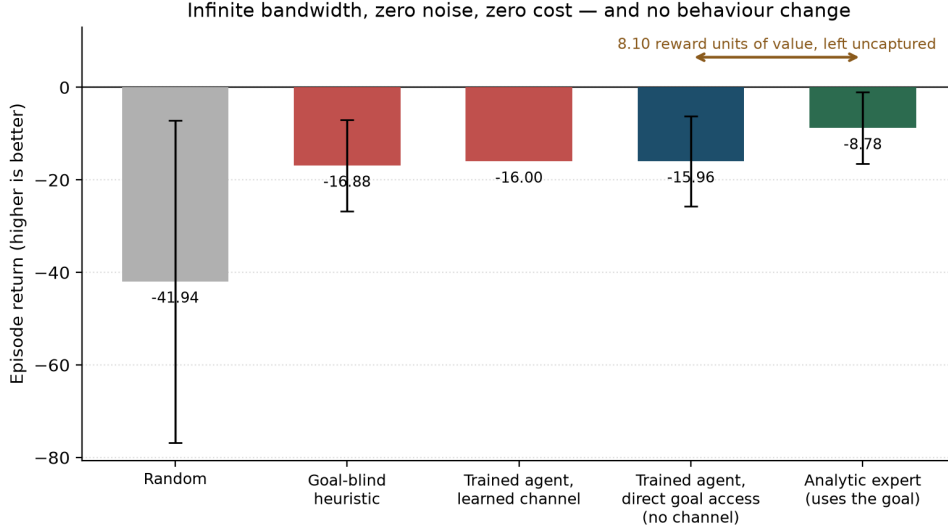


Figure 5: Removing the channel entirely and handing the receiver the sender’s goal directly does not move behaviour off the goal-blind optimum, even though an analytic policy shows 8.10 reward units are on the table for using it.

block-shuffled surrogate correction removed essentially the entire effect. We report this because it illustrates the paper’s thesis at the level of practice: observational coupling measures can produce confident, well-behaved, theoretically congenial numbers in the complete absence of coupling. It also stands as a concrete instance of the estimator fragility documented independently for inter-brain synchrony [43].

7 Why the gap exists

The dissociation is not an artefact of weak optimisation, and understanding why illuminates the general case.

Coupling is not a property of a channel. It is a jointly held convention, and conventions are attractors of a shared history rather than consequences of contact. Two facts follow.

Coupling cannot be established unilaterally. Information has value to a system only if that system already possesses machinery that acts on it; acquiring that machinery is rewarded only if the information is already being acted upon. Each half is a precondition for the other, so neither party can move first. Meanwhile there is always a policy available that ignores the partner and performs moderately well alone — in our system, hedging towards the centroid of the candidate landmarks. That policy is reachable immediately, unilaterally, without coordination. It is found early and it is stable: viewed from inside it, the partner’s signal is worth nothing, so no local gradient points out of it.

Our agents fell into precisely this attractor and never left, with the answer sitting in their own hidden layer the entire time. We tested a series of independent escapes — longer training, learning-rate and gradient-batching sweeps, exploration pressure, an information-aware value baseline, and auxiliary reconstruction objectives on the sender and on the receiver — and none produced behavioural change, because each optimises *within* the basin rather than across its boundary. Section B tabulates them. The last two are the sharpest: they repaired both halves of the state-level pipeline, raising encoding fidelity to $R^2 = 0.90$ and driving the receiver’s reconstruction error to 0.0017, and behaviour did not follow.

Coupling should therefore be bistable, and hence historical. If coupling is an attractor rather than a channel state, the conditions under which it ignites need not match the conditions under which an established convention collapses. We predict hysteresis: a dyad that has already established a convention will sustain coupling under degradation that would have prevented its formation. If this holds, coupling capacity is not a function of a dyad’s present parameters at all, only of its history — and cross-sectional measurement of coupling capacity is ill-posed. This also explains an everyday asymmetry: establishing a shared code, whether a language, a jargon or a couple’s private shorthand, is enormously expensive, while maintaining one is nearly free. [Open Question 3](#) states the experiment that would decide it.

8 Interventional coupling measurement

The diagnosis prescribes the treatment. Functional coupling is defined interventionally, is not identified observationally, and must therefore be estimated by intervention.

Protocol. To establish that a signal is doing work, ablate or randomise it and measure the change in the receiver’s behaviour. Functional coupling is what survives removal of the signal.

8.1 Three tiers of intervention

1. **Ablation.** Replace the signal with a null or constant and measure the behavioural difference. Establishes whether the channel carries any functional load.
2. **Randomisation.** Replace the signal with one drawn from the same marginal but paired with a different underlying state. This is the decisive test against the confound of [Theorem 6](#): it preserves every surface property of the signal while destroying its relationship to the sender’s state. A receiver acting on shared prior convention rather than on the message is unaffected; a genuinely coupled receiver degrades.
3. **Targeted perturbation.** Alter the signal so as to change its meaning under the putative convention while holding surface statistics fixed, and check that behaviour tracks the altered meaning. Establishes not only that the channel matters but that it matters *in the way the convention specifies*.

Randomisation deserves emphasis: it is what defeats $\mathcal{D}_2$. Under $\text{do}(M = m')$ with m' drawn independently of C , the coupled dyad’s behaviour changes and the confounded dyad’s does not, which is exactly the discrimination [Theorem 6](#) shows observation cannot make. [Section 6](#) confirms empirically that it is also the more sensitive of the two, detecting residual dependence that ablation reports as null.

8.2 A scale: the captured share of the value of information

An interventional test still requires a *scale*. Establishing that behaviour changes when a signal is removed does not establish that the signal is doing appreciable work; our own receiver is a case in point. We therefore propose reporting the effect as a fraction of what the information is worth.

Definition 21 (Captured share of the value of information). Let $J(\pi)$ be the receiver’s expected task performance under policy π . Let π_{inf}^* be an optimal policy with access to the sender’s state

and π_{blind}^* an optimal policy without it. The *value of information* is

$$\text{VoI} = J(\pi_{\text{inf}}^*) - J(\pi_{\text{blind}}^*) \geq 0,$$

and the captured share achieved by the observed receiver π is

$$\rho(\pi) = \frac{J(\pi) - J(\pi_{\text{blind}}^*)}{\text{VoI}}, \quad \rho \in (-\infty, 1].$$

Three properties make ρ the right headline number. It is dimensionless, so it is comparable across tasks with different reward scales. It is zero exactly when the receiver does no better than the best policy that ignores the signal, which is the substantive null — rather than “the intervention changed nothing,” which is a weaker and easier-to-satisfy null. And it is bounded above by 1, so a small ρ cannot be rescued by a large sample: statistical significance and practical magnitude are separated by construction.

Estimation. $\hat{\rho}$ is a ratio of differences of means and is straightforward to estimate: evaluate each of the three policies on a common set of n episodes or trials, and obtain a confidence interval for $\hat{\rho}$ by a nonparametric bootstrap over episodes, resampling the numerator and denominator jointly so that shared task variation cancels. Where the informed optimum is genuinely unavailable, the uninformed optimum alone still provides a far more meaningful reference than zero, and the numerator $J(\pi) - J(\pi_{\text{blind}}^*)$ can be reported in task units.

Anchors. The two anchors need not be learned. In our environment both are analytic (Table 3). In general: π_{blind}^* can be obtained by training or hand-designing a policy that is architecturally denied the signal, and π_{inf}^* by an oracle policy or a human ceiling. In discrete evaluation settings the analogues are the base rate of satisfying an unstated constraint by chance, and the compliance rate of a response constructed with the constraint made maximally salient (Section C).

8.3 Reporting checklist

We suggest that any claim of coupling — between agents, between brains, or between a user and a model — report the following. Each item corresponds to a failure documented in this paper.

1. The intervention used, from the three tiers above; randomisation at minimum (Section 6).
2. $\hat{\rho}$ with a bootstrap interval, alongside the raw effect (Definition 21).
3. Both anchors, and how they were obtained (Table 3).
4. For any instrumented estimate, the first-stage F -statistic (Section 5.2; 37), and Balke–Pearl bounds when it is marginal (Section 5.4).
5. For any observational coupling statistic, a surrogate or pure-noise control establishing the estimator’s bias at the sample size used (Remark 20).
6. An explicit statement of whether C was observed, proxied, or unobserved (Assumption A2).

9 Implications

9.1 Evaluation of AI systems

By Corollary 11, conversational and preference-based evaluation cannot identify functional coupling between a model and a user’s intent. This is a structural limitation, not a matter of

benchmark quality or rater training. The prescription is concrete: evaluate by intervention. Perturb the user’s signal in ways that change intent while preserving surface form, and measure whether model behaviour tracks the change. A system whose outputs are invariant to such perturbations is exhibiting $\mathcal{D}_2$ — coherent, fluent and uncoupled. [Section C](#) gives a complete pre-registered design.

We note the practical stake. Apparent and functional coupling produce identical behaviour *within* the distribution on which apparent coupling was established, and divergent behaviour outside it. Our receiver looked unremarkable — it scored what every other agent scored — until the situation actually required the signal, at which point it failed completely. On this account a familiar deployment pattern, in which systems pass extensive evaluation and then fail in the field, is not a symptom of insufficient testing but of testing a quantity that does not determine the one that matters.

9.2 Mechanistic interpretability

A claim that a model represents some quantity is a statement about decodability from internal state — precisely the measure that read 0.0017 in [Table 2](#) while behaviour read effectively zero. Presence of a feature in a representation licenses nothing about its causal role. The field’s causal methods — activation patching, causal scrubbing and related interventions [\[8\]](#) — are therefore not refinements of decoding-based analysis but the only valid form of the claim. Our result is a clean existence proof of a feature that is fully present, linearly recoverable and behaviourally inert.

[Corollary 18](#) sharpens this into a claim about the field’s trajectory. Because the obstruction is structural, it is not relieved by better probes, larger probing datasets or finer-grained feature decomposition: a decoding result of arbitrarily high fidelity still licenses no behavioural conclusion. Progress on decoding accuracy and progress on causal warrant are therefore orthogonal, and effort spent on the former should not be scored as progress on the latter.

Neuroscience reached a structurally identical conclusion earlier and with a different vocabulary. Grootswagers et al. [\[10\]](#) show with MEG that category information is broadly decodable across visual cortex while only a subset of it predicts trial-by-trial categorisation behaviour, and the motor literature has built a mature framework around exactly this distinction, in which a large fraction of population activity occupies an output-null subspace that cancels at the behavioural readout [\[15\]](#). That these fields converged independently on the same distinction is evidence that the gap in [Table 2](#) is a general property of layered systems rather than a quirk of our training setup.

9.3 Hyperscanning and inter-brain coupling

Inter-brain synchrony and directed-information measures are observational, state-level, and applied to dyads that share both a task and a culture — that is, a large common cause. Three of our results bear on this simultaneously, and their conjunction is unusually restrictive. [Theorem 6](#) applies directly. [Proposition 16](#) closes the most natural escape: transfer entropy and Granger causality are the field’s standard instruments precisely because they are directional, but directionality obtained from temporal ordering does not survive a common cause that precedes both series, and shared convention is exactly such a cause. [Proposition 17](#) closes the second: recording more of the receiving brain, at higher resolution, does not help, because the failure is structural rather than one of measurement precision. And [Proposition 19](#) adds that even were confounding absent, the quantity being measured is at the wrong node. Reported inter-brain coupling therefore establishes that two brains’ states covary; it does not establish that either

brain’s behaviour depends on the other’s signal.

The field has independently been moving in a compatible direction, which sharpens rather than weakens the point. Varlet and Grootswagers [41] show that inter-brain synchrony has low sensitivity to whether two people are actually processing the same information, and propose cross-brain representational analysis as a replacement; Markus and Shamay-Tsoory [21] name the directionality problem explicitly and propose dyadic neurofeedback, Granger causality and structural equation modelling as remedies; and Zimmermann et al. [43] show that arbitrary preprocessing choices skew synchrony estimates substantially. Our results imply that the first two of Markus and Shamay-Tsoory’s three remedies are statistical and therefore remain under Theorem 6 and Proposition 16, while the third — real-time neurofeedback — is genuinely interventional and is the one to pursue. The commonly used pseudo-pair control is a step in the same direction but controls for the shared *stimulus* rather than for the coupling channel itself, and so does not reach $\mathcal{D}_2$.

There is also a constructive path that requires no new experiments. By Remark 15, exogenous variability in the transmitted signal — momentary occlusion, acoustic interference, dropped words, latency — is an instrument, provided it is independent of the pair’s shared history. Existing hyperscanning corpora contain such variation and typically treat it as noise to be cleaned. Recovering it and using it as an instrument is, to our knowledge, an unexploited route to interventional conclusions from already-collected observational data — subject to the first-stage diagnostic that our own Stage-2 attempt failed (Section 5.2). We further suggest that inter-brain studies adopt signal randomisation as a standard control, and note a testable consequence: on a fixed task with fixed instrumentation, measured coupling should track the pair’s *prior shared convention* — strangers below acquaintances below long-term partners — and that effect should exceed any channel-quality variable.

9.4 Brain-to-brain interfaces

Direct neural interfaces are motivated by a bandwidth argument, and its premise is well supported. Language is a narrow, serial channel. Across seventeen languages spanning nine families, conversational speech converges on an information rate near 39 bits/s, syllable-rate differences being compensated almost exactly by differences in information per syllable [4]; the throughput of deliberate human behaviour more generally has been estimated at roughly 10 bits/s, some eight orders of magnitude below the raw capacity of the sensory periphery [42]. The inference usually drawn is that the mouth and the ear are the binding constraint on what two minds can accomplish together — that cognition is fast and parallel while its expression is slow and serial, and that a cortical link bypassing articulation and audition would let brains exchange at the rate at which they compute.

The present account grants the premise and rejects the inference. The rate limit is real and is not what binds. Section 6.3’s infinite-bandwidth control is this hypothesis tested in the only setting where it can be tested cleanly: we removed the channel entirely and delivered the sender’s private state to the receiver directly, noiselessly, at no cost — the limit that no physical interface can exceed — and behaviour did not move. Capacity was never the quantity in short supply. A pair that does not share a convention gains nothing from a wider channel, because the gain from widening a channel is bounded by the amount of the signal the receiver was going to act on, and that amount was approximately zero.

The empirical record on real brain-to-brain systems agrees. The framework retrodicts a standing anomaly, and the anomaly is documented rather than asserted. Rao et al. [29] (2014; 64-channel EEG sender, one TMS coil, three pairs) transmitted 0.25–0.81 bits/trial. Jiang et al. [13] (2019; a real-time, three-person, two-sender architecture, the field’s most technically advanced

system) transmitted 0.336 bits/trial for its best sender — *less* than Rao’s best pair five years earlier, despite strictly more sophisticated hardware and with only a single electrode actually used for decoding out of the 8- and 32-channel systems available. The authors state this themselves: “From the first human BBI to BrainNet, the level of information complexity has remained binary... [and] this low bit rate required a disproportionate amount of technical hardware and setup” [13]. An independent systematic review of the 2013–2020 literature [25] reports the same pattern across the full set of published studies: methodological diversification (EEG, intracortical microelectrodes, focused ultrasound, optogenetics; TMS, direct cortical stimulation) without a corresponding rise in bits transmitted, concluding that mutual information “has been limited to mostly binary information transfer” throughout.

On a channel-centric account this is unexplained, and worsens with better hardware rather than improving. On the present account it is forced: in every such study the convention is installed from outside, in advance, by ordinary language — the participant is *told* that a phosphene means “rotate,” or the meaning is fixed by the experimenter’s choice of stimulation site. Throughput is therefore bounded not by hardware but by how much convention can be pre-installed by talking to a subject beforehand. That bound is low and does not move when the hardware improves; if anything, coordinating a second sender in BrainNet added convention-installation overhead without adding transmitted content.

This converts into a design claim rather than only a retrodiction. If throughput is bounded by how much convention can be installed in advance, the way to raise it is to make convention formation part of the interface instead of a precondition the experimenter supplies verbally beforehand: bidirectional, closed-loop, and trained across sessions, so that the pair acquires a code neither party held at the outset. [Section 7](#) argues that this is a joint exploration problem and therefore expensive — neither side can construct the code unilaterally, which is one reason a child spends years acquiring the conventions of a first language rather than minutes acquiring its channel. The prediction is that a direct interface does not avoid that cost; it relocates it into a code for which neither brain has any prior, and better electrodes do not pay it. [Open Question 4](#) states the corresponding falsifier.

10 Related work

The components of this argument exist; the synthesis and the identifiability claim, to our knowledge, do not.

Foundations. Shannon [33] explicitly excluded semantics from the engineering problem, a simplification that was safe while receivers executed protocols fixed in advance and fails once both ends must acquire their protocol. Pearl [27] supplies the ladder on which we locate functional coupling. Searle [32] argued that syntactic competence does not constitute understanding but offered no test; our contribution can be read as supplying an operational criterion and explaining why conversational tests could never have provided one. Lewis [18] formalised convention; Harnad [11] posed symbol grounding; Skyrms [34] derived meaning from signalling games.

Directed information and its use between systems. Schreiber [30] introduced transfer entropy for model-free asymmetric dependence between time series and Massey [22] formalised directed information for systems with feedback; Kraskov et al. [17] supply the k -nearest-neighbour estimators used as a model-free cross-check in [Section 2.3](#). These are the tools from which [Definition 1](#) was assembled, and [Proposition 16](#) is our statement of what they cannot do between two systems that share a convention. On the interface side, Pais-Vieira et al. [26], Rao et al.

[29] and Jiang et al. [13] established physical feasibility of direct neural links under severe bandwidth constraints; Thual et al. [39] align functional cortical geometries across subjects by optimal transport, and Nakamura et al. [24] perform zero-shot representation transfer with hyperspherical embeddings. On our reading, alignment is a technique for raising the quantity in Definition 1, not for identifying $\mathcal{F}$, and Section 9.4 argues that the constraint these systems face is not the one they are engineered against. Friston and Frith [7] model communication as mutual free-energy minimisation and Stephens et al. [36] document speaker–listener neural coupling; both motivate why coupled systems synchronise without supplying a criterion that separates $\mathcal{D}_1$ from $\mathcal{D}_2$. Self-supervised neural encoders — LaBraM [14], BrainLM [3], MindEye2 [31] — are the state extractors on which a biological instantiation of these measures would be built, and by Corollary 18 their improvement does not change the identifiability situation.

Nearest formal neighbour. Kolchinsky and Wolpert [16] define semantic information as information with causal relevance to a system’s viability. Like us, they locate meaning in causal consequence rather than correlation. Unlike us, they do not address identifiability from observational data or the constitutive role of shared convention as confounder, and their formalism is monadic — one system’s information about its environment — rather than dyadic, which is the structural reason it does not already reach the coupling question we ask. Their contribution is a definition; ours is an impossibility result about recovering a quantity of that kind from observation.

Emergent communication. Lowe et al. [19] distinguish positive signalling from positive listening and warn that the former can be high while the latter is absent — a distinction we rediscovered empirically before encountering their terminology, and which Table 2 instantiates with ground truth. Their matrix-communication-game experiments give an independent replication in a different environment: positive signalling (their Speaker Consistency, 0.19–0.54 across game sizes) coexists with causal influence at the floor, message removal leaving action-prediction accuracy essentially unchanged. They trace this to an architectural confound — shared hidden layers between action and communication heads induce spurious signalling correlations even with untrained communication parameters. Our diagnostic chain rules out that mechanism for our system (separate networks, an auxiliary loss raising encoding fidelity to $R^2 = 0.90$, and a listener reconstruction head decoding the goal at error 0.0017) and the dissociation persists regardless — the same symptom via a different route, which is why we read their result as a second instance of the phenomenon rather than an explanation of ours. They frame the finding as a metric pitfall with a practical remedy (measure causal influence, use several estimators); we argue no estimator escapes it without intervention. Eccles et al. [5] introduce explicit biases to induce both signalling and listening, motivated by the joint-exploration failure Section 7 describes; Foerster et al. [6] address the same obstacle architecturally.

Decodable but unused. The distinction between information that is present and information that is used is well established within single systems: Grootswagers et al. [10] give a direct human-neuroscience instance, and Kaufman et al. [15] give the mature theoretical vocabulary (output-null versus output-potent subspaces). Our contribution is to carry the distinction *across* systems, where it interacts with the convention confounder, and to show that no observational measure recovers it.

Hyperscanning methodology. Varlet and Grootswagers [41] argue that inter-brain synchrony is insensitive to information alignment and propose representational analyses instead; Markus

and Shamay-Tsoory [21] frame the problem as one of directionality; Zimmermann et al. [43] document estimator fragility. These converge on distrust of raw synchrony from three different directions. None frames the problem as non-identifiability, and none proposes channel ablation or randomisation, which is where our account differs.

Interpretability. Interpretability’s causal turn [8] already reflects awareness that decoding is insufficient; we supply an argument for why intervention is necessary rather than merely preferable, and, via [Corollary 18](#), why the necessity does not diminish as probes improve.

11 Limitations and falsification

We state the ways this could be wrong, in order of severity.

The superseded programme is reported, not defended. [Section 2](#) states the coupling-capacity formalism and [Theorem 2](#) because they are the origin of the argument and because we take the theorem to be correct as a bound on what a bandwidth-limited interface can drive. Its empirical support is nonetheless weak in the specific way [Section 2.3](#) sets out, and the quantity it bounds is exactly the quantity [Theorem 6](#) shows to be unidentifiable. A reader assessing the paper’s positive claims should weigh [Sections 4 to 6](#) and [8](#); [Section 2](#) is included for the record, and because the failure it describes is what the rest of the paper explains. Readers of the earlier manuscript should treat its headline validation claims as withdrawn in the form in which they were stated there.

The assumption set may be narrowable further. [Theorem 13](#) gives three sufficient conditions and [Propositions 16](#) and [17](#) rule out two tempting shortcuts, but we do not claim this characterisation is exhaustive. A weaker assumption set that still identifies $\mathcal{F}$ — particularly one satisfiable in observational hyperscanning corpora, where intervention is impossible after the fact — would be a substantial improvement and is where we would look first to strengthen the paper.

The instrumental route works on the construction built to demonstrate it and fails on the first system we did not build for it. Applying [Remark 15](#)’s method to the trained speaker–listener, using the channel’s own transmission noise as instrument, gives an unusable estimate (first-stage $F = 0.2$; [Section 5.2](#)) because the auxiliary loss that makes the channel informative also drives it towards confident, low-noise-sensitivity outputs. This is a real, diagnosed boundary condition rather than a refutation of the method — the first-stage check caught it correctly rather than reporting a misleading point estimate — but it means the route is not yet shown to work on any system we did not design to make it work, and whether it survives contact with real, non-optimised biological channel noise remains open ([Open Question 2](#)).

The empirical demonstration is a single system. [Proposition 19](#) is an existence proof, which is what it needs to be, but it does not establish that the gap is common in systems of practical interest. If ablation and observational measures agree across a broad range of real dyads, the correction we prescribe is unnecessary in practice even if valid in principle.

Functional coupling here is small, not provably zero. The randomisation effect ($z = +1.82$) is marginal by conventional standards and we do not claim it is null; a larger evaluation set would sharpen it in one direction or the other. Our claim is the weaker and better-supported one that the receiver captures almost none of the available value of the information while every

state-level measure is at ceiling. Readers should note that this makes the dissociation a matter of degree, and that our recommended statistic — $\hat{\rho}$ — is what carries the argument, not a significance test.

Hysteresis is predicted, not shown. Section 7’s bistability claim is untested. Coincident onset and collapse thresholds would falsify it and restore coupling to the status of a state variable of the connection.

The retrodiction rests on few data points. Section 9.4 cites measured bits/trial from the two best-documented studies (2014 and 2019, five years apart, throughput flat-to-declining despite more sophisticated hardware) plus an independent systematic review’s qualitative agreement, rather than asserting the pattern from memory. But these are two quantified anchor points bracketing a decade, not a dense time series, and a general search of the 2021–2026 literature found no further quantified human-to-human study either confirming or breaking the pattern — an absence of evidence, not evidence of the pattern’s continuation. A more exhaustive database search, or a new study reporting a materially higher bit rate, would each bear directly on this claim.

Scope. We address functional coupling as defined in Definition 5 and make no claim about phenomenal understanding, consciousness, or meaning in a richer sense. A reader who holds that understanding is not exhausted by causal influence on behaviour should read our results as necessary but not sufficient conditions.

12 Open questions

The account above is complete as an argument and incomplete as a research programme. We state the outstanding questions in the form in which we intend to pursue them, ordered by how much a determinate answer would change the paper. Each is posed so that both answers are informative.

Open Question 1 (Does the dissociation occur in deployed conversational systems?). Remark 10 argues that a language model in conversation is a compact description of $\mathcal{D}_2$. That is currently an argument, not a measurement. The test is to hold a request byte-identical while varying a previously stated constraint, and to compare intact, ablated, randomised and targeted-perturbation conditions on the ρ scale of Definition 21. A finding that behaviour tracks the specific substituted constraint would narrow the claim — the theory says apparent and functional coupling *can* dissociate and are not separable by observation, not that every exchange is $\mathcal{D}_2$ -like — while a finding that responses remain fluent and on-topic but constraint-invariant would convert Section 9’s prescription from speculation into a reported result. Section C gives the full design, including the confound (context-window recency mechanics) that must be separated from the effect of interest before either answer can be believed.

Open Question 2 (Does the noise-as-instrument route survive real biological channels?). The method works on a controlled construction and fails on a trained discrete channel, for a diagnosed reason (Section 5.2). The open question is whether the failure mode is specific to optimised, saturated codes or general. Speech and neural channels are not trained to be reliable through a threshold, so we expect stronger instruments there, but this is a prediction. The concrete test is to identify an exogenous variability source in a public hyperscanning corpus, argue explicitly for its independence from the pair’s shared history, and report the first-stage F before any estimate.

A weak instrument there too would mean the observational route is closed in practice as well as in principle, leaving only [Theorem 13\(iii\)](#).

Open Question 3 (Is coupling bistable?). [Section 7](#) predicts hysteresis: the channel degradation at which an established convention collapses should be strictly worse than the degradation that would have prevented its formation. The experiment is a bandwidth or noise sweep run twice on the same dyad, once ascending from an uncoupled start and once descending from an established convention. Coincident thresholds falsify the attractor account and restore coupling to a state variable of the connection; separated thresholds imply that cross-sectional measurement of coupling capacity is ill-posed, which would be the more consequential of the two answers.

Open Question 4 (Can a bidirectional interface raise throughput above the pre-installed-convention bound?). [Section 9.4](#) predicts that brain-to-brain throughput is bounded by how much convention can be installed verbally in advance, and that hardware improvements do not move that bound. The falsifier is direct: a closed-loop, bidirectional, multi-session interface in which a pair acquires a code neither held at the outset and exceeds the flat $\lesssim 1$ bit/trial ceiling of the existing literature. Success would confirm the design claim; failure at high channel quality would strengthen the retrodiction; and success achieved *without* multi-session co-adaptation would falsify the account outright.

Open Question 5 (How large is the gap between $\mathcal{F}$ and its observational proxies in practice?). [Proposition 19](#) is an existence proof. It says nothing about prevalence. The useful quantity is the empirical distribution, across many dyads and tasks, of the discrepancy between an observational coupling statistic and the interventional estimate on the same data. If that discrepancy is usually small, the theory stands and the practical correction is minor; if it is usually large, the observational literature requires re-examination rather than re-labelling. We know of no dataset on which both quantities have been computed.

Open Question 6 (Is there an identifying assumption weaker than [Theorem 13\(i\)–\(iii\)](#)?). Hyperscanning corpora are already collected and cannot be intervened upon retrospectively. Any assumption that identifies $\mathcal{F}$ — or usefully bounds it beyond [Section 5.4](#) — from a fixed observational corpus would be worth more to that field than everything else in this paper. Negative sharpening is also valuable: a proof that no such assumption exists within a stated class would convert [Theorem 6](#) from an impossibility for current methods into an impossibility for the setting.

Open Question 7 (What is the right notion of convention distance?). Several predictions above — that measured coupling tracks strangers below acquaintances below long-term partners, that adjustment on a proxy for C yields partial control — presuppose a way of quantifying how much convention a dyad shares. We do not have one. A defensible measure of convention distance, validated against behavioural coordination, would make [Theorem 13\(i\)](#) approximately usable and would turn several qualitative predictions into quantitative ones.

13 Conclusion

Four fields ask whether one system understands another, and all four answer by observing the exchange. We have argued that this cannot succeed. Shared convention is the precondition of communication and simultaneously a common cause of what is sent and how the receiver acts, so genuinely coupled and merely coherent dyads can be observationally identical — not as a knife-edge construction but on an open set of parameters. Independently of that confound, a receiver’s internal state can contain a signal in full while its behaviour extracts almost none of

its value. We demonstrated the latter in a system where the sender encoded at $R^2 = 0.90$, the receiver’s hidden layer carried the content to a reconstruction error of 0.0017, and the receiver nonetheless performed at the goal-blind optimum, capturing approximately 12% of what the information was worth — and where handing it that information directly, with infinite bandwidth and no noise, changed nothing at all. Along the way we found that the receiver’s behaviour was faintly perturbable by the signal while deriving no advantage from it, so that sensitivity and benefit dissociate in their turn, and an interventional test needs a scale rather than a null.

We arrived at this by trying to build the opposite. [Theorem 2](#) still says what a bandwidth-limited interface can drive into a receiver of finite effective dimension, and we expect it to be useful to anyone designing one. But a bound on what a channel could deliver is not a measurement of what a receiver does with it, and the quantity that bound governs is a functional of the observational distribution, so it inherits everything above.

Functional coupling is an interventional quantity. It has been measured observationally. That, rather than any deep obscurity in the phenomenon, is why the question of whether one mind understands another has resisted answer: the instrument was on the wrong rung. The remedy is available and cheap. Remove the signal, and see what changes.

Data and code availability

All code, configurations, random seeds, logged results and figure-generating scripts accompany this manuscript. The identifiability demonstrations are in `experiments/paper2_identifiability/` (`noise_as_instrument.py` for the controlled construction of [Proposition 8](#), `noise_as_instrument_stage2.py` for the trained-system application). The trained speaker–listener system, its four escape attempts and the oracle control are in `experiments/paper1_rl/`, whose `run_experiments.py` also produces the superseded ground-truth simulation and the panels of [Fig. 1](#). Numerical results are logged as JSON under `experiments/results/logs/`. The study-by-study brain-to-brain throughput tabulation supporting [Section 9.4](#) is in `literature/bbi_throughput_survey.md`.

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Appendix A Proof of Proposition 8

All variables are zero-mean and jointly Gaussian, so the joint law of (M, U) is determined by $\text{Var}(M)$, $\text{Var}(U)$ and $\text{Cov}(M, U)$. Throughout, $M = C + N$ with $C \perp\!\!\!\perp N$, hence $\text{Var}(M) = \sigma_C^2 + \sigma_N^2$.

Covariance matching. In $\mathcal{D}_1$, $\text{Cov}(M, U) = \kappa \text{Var}(M) = \kappa(\sigma_C^2 + \sigma_N^2)$. In $\mathcal{D}_2$, $\text{Cov}(M, U) = \kappa' \text{Cov}(C, C + N) = \kappa' \sigma_C^2$. Equating gives $\kappa' = \kappa(\sigma_C^2 + \sigma_N^2)/\sigma_C^2$, as stated.

Variance matching. In $\mathcal{D}_1$, $\text{Var}(U) = \kappa^2(\sigma_C^2 + \sigma_N^2) + \sigma_\varepsilon^2$. In $\mathcal{D}_2$, $\text{Var}(U) = \kappa'^2 \sigma_C^2 + \sigma_{\varepsilon'}^2 = \kappa^2(\sigma_C^2 + \sigma_N^2)^2/\sigma_C^2 + \sigma_{\varepsilon'}^2$. Equating and solving,

$$\sigma_{\varepsilon'}^2 = \sigma_\varepsilon^2 + \kappa^2(\sigma_C^2 + \sigma_N^2) \left[1 - \frac{\sigma_C^2 + \sigma_N^2}{\sigma_C^2} \right] = \sigma_\varepsilon^2 - \kappa^2 \frac{(\sigma_C^2 + \sigma_N^2) \sigma_N^2}{\sigma_C^2},$$

which is a valid variance precisely when (5) holds. Since $\text{Var}(M)$ is unchanged between the dyads by construction, all three moments agree and $P_1(M, U) = P_2(M, U)$.

Interventional separation. In $\mathcal{D}_1$, $\text{do}(M=m)$ gives $U = \kappa m + \varepsilon$, so $\mathbb{E}[U \mid \text{do}(M=m)] = \kappa m$ varies with m and $\mathcal{F}(\mathcal{D}_1) > 0$ for $\kappa \neq 0$. In $\mathcal{D}_2$, U is generated from C and ε' alone, so $\text{do}(M=m)$ leaves the law of U unchanged and $\mathcal{F}(\mathcal{D}_2) = 0$.

Openness. Condition (5) is a strict inequality between continuous functions of $(\kappa, \sigma_C^2, \sigma_N^2, \sigma_\varepsilon^2)$ on the domain $\sigma_C^2, \sigma_N^2 > 0$, so the set of parameter vectors satisfying it is open and non-empty (take σ_ε^2 large). The equivalence therefore does not depend on exact tuning. $\square$

Corollary for the naive estimator. The population OLS coefficient of U on M is

$$\frac{\text{Cov}(M, U)}{\text{Var}(M)} = \begin{cases} \kappa & \text{in } \mathcal{D}_1, \\ \frac{\kappa' \sigma_C^2}{\sigma_C^2 + \sigma_N^2} = \kappa & \text{in } \mathcal{D}_2, \end{cases} \quad \text{whereas} \quad \frac{\text{Cov}(U, N)}{\text{Cov}(M, N)} = \begin{cases} \kappa & \text{in } \mathcal{D}_1, \\ 0 & \text{in } \mathcal{D}_2. \end{cases}$$

The naive regression therefore returns the same value in both dyads, and it is the correct interventional effect in exactly one of them — with nothing in the data to say which. The instrumental estimator separates them, which is what Section 5.2 verifies numerically.

Appendix B Experimental detail

B.1 Environment and agents

The task is PettingZoo’s `simple_speaker_listener` [38], in which a stationary speaker observes which of three landmarks is the goal and a mobile listener, which cannot see the goal, must navigate to it. Episodes run 25 steps; the return is the negative cumulative distance from the listener to the goal landmark. The environment’s built-in fixed-vocabulary communication channel is bypassed.

In its place, the speaker emits an n -bit message through a straight-through Gumbel–Sigmoid channel [12] of tunable width, so that gradients from the listener’s loss reach the speaker’s parameters [6]. Speaker and listener are separate multilayer perceptrons with no shared parameters — a deliberate choice, since Lowe et al. [19] identify shared hidden layers between action and

communication heads as a source of spurious signalling correlations, and we wished to exclude that mechanism.

B.2 Training configuration

The configuration reported throughout is: bandwidth $B = 8$ bits/step; 20,000 episodes; Adam at $\eta = 5 \times 10^{-4}$; 16 episodes per gradient update; entropy coefficient 0.02; speaker auxiliary reconstruction coefficient 200; listener auxiliary reconstruction head enabled; seed 42. Evaluation uses 150 held-out episodes with frozen, detached policies.

Two configuration details are worth recording because they cost time. First, the learning rate is not free: 3×10^{-3} diverges, and 5×10^{-4} is the settled value. Second, the auxiliary coefficient must be large: a `reduction="sum"` versus `"mean"` scale mismatch made `aux_coef = 1.0` negligible, and encoding fidelity moved from $R^2 = 0.001$ to $R^2 = 0.90$ only at `aux_coef  $\approx$  200`. A message-aware value baseline that was not detached reproduced the same trap in the opposite direction, regressing encoding fidelity from 0.90 to 0.0003.

B.3 Escape attempts

Section 7 claims that the goal-blind policy is an attractor rather than an artefact of insufficient optimisation. Table 4 records the interventions tested. Each was well motivated; each optimises within the basin.

Intervention	Outcome
Longer training (3k, 4k, 20k episodes)	No effect on coupling
Learning-rate tuning	3×10^{-3} unstable; 5×10^{-4} settled. No effect on coupling
Gradient batching (1/16/64 episodes per update)	Improves task return substantially; <i>zero</i> effect on encoding
Exploration pressure (entropy coefficient 0.02/0.005/0)	No effect on anything measured
Message-aware value baseline	No effect when detached; regresses encoding when not
Listener auxiliary reconstruction head	Fixes the <i>representation</i> (error 0.0017), not the behaviour
Speaker auxiliary loss	Works on <i>encoding</i> ($R^2 = 0.90$); no behavioural change

Table 4: Interventions tested against the goal-blind attractor. The last two rows are what produced the state-behaviour gap of Table 2: both halves of the state-level pipeline were repaired, and behaviour did not follow.

B.4 Standing analysis rules

Two rules were adopted after the artefact described in Remark 20 and are applied to every number reported here. No transfer-entropy estimate is reported without a block-shuffled surrogate correction; and the evaluation sample count is confirmed to exceed the channel dimensionality by a comfortable margin before any information-theoretic quantity is computed. Remark 20 is what happens when these are skipped.

Appendix C Pre-registered design for the human–AI dissociation study

This appendix records the design for [Open Question 1](#) in advance of running it, so that the analysis is fixed before the data exist. It is the direct analogue of [Section 6](#) on a deployed conversational system, and is deliberately cheap: no specialised hardware, no participant recruitment.

C.1 Task structure

Every task is a two-turn exchange. Turn 1 states an arbitrary, task-relevant private constraint — a dietary restriction, a coding-style preference, a target-audience level, a hidden budget, a required tone. Turn 2 is a request that *should* be shaped by turn 1 but whose literal text is held **byte-identical** across every constraint variant within a task family. Holding turn 2’s surface form fixed is what makes this an intent-versus-surface test: the only thing that varies is what turn 1 said, never how turn 2 is phrased.

Five families are specified, with at least six constraint variants each: recipe (dietary restriction), code style, explanation level (audience), budget, and tone. Compliance is defined per family before any data is collected.

C.2 Conditions

The four conditions mirror [Section 8](#) exactly.

1. **Intact** — real turn 1 and real turn 2. Establishes baseline compliance.
2. **Ablation** — turn 2 alone. Establishes whether turn 1 does any work at all.
3. **Randomisation** — turn 1 replaced by a different constraint from the same family’s pool, matched in position, length and register, then the real turn 2. This is the sharp test: does the output track *which* constraint was stated, or would any turn-1-shaped text produce the same downstream behaviour? A system exhibiting $\mathcal{D}_2$ produces a response indistinguishable from intact — fluent, on-topic, and not keyed to the substituted content.
4. **Targeted perturbation** — turn 1’s surface form held near-identical with only the specific value flipped. Strongest test: does the specific content of the output flip to match?

C.3 Scoring and statistics

Responses are scored against a narrow rubric written before data collection, one rubric per family. A blind subsample is hand-rated and inter-rater agreement reported regardless of how it comes out. Scoring records not only whether the response satisfies *some* constraint but whether it differs from the intact-condition response in a way that tracks the substituted content — a response that is identical regardless of which constraint was stated exhibits zero functional coupling even if it happens to comply by accident. If the system explicitly flags the mismatch, that is evidence *of* functional coupling (behaviour changed under the intervention) and is scored as such rather than folded into non-compliance.

The headline statistic is $\hat{\rho}$ ([Definition 21](#)), with the informed anchor given by a response generated with the constraint made maximally salient and the blind anchor by the base rate of satisfying a randomly drawn constraint by chance. Standardised effects between intact and each intervention condition are reported alongside $\hat{\rho}$, never instead of it.

C.4 Confounds fixed in advance

Three are pre-specified. *Recency mechanics*: a system might fail to track turn 1 because of an architectural recency bias rather than the confound of [Theorem 6](#). The distance manipulation — inserting 0, 2 or 5 filler turns between turn 1 and turn 2 — separates these, and both are reported if they diverge. *Judge bias*: mitigated by the hand-rated subsample above. *Ecological validity*: these are two-turn exchanges, not deployed usage, and any writeup says so. *Single-system scope*: whichever system is tested, the result is an existence proof for that system, exactly as [Proposition 19](#) is for ours.

C.5 Interpretation, fixed in advance

If the randomisation and targeted-perturbation conditions track the substituted content, that is genuine functional coupling in that system and task regime, and it *narrows* the paper’s claim rather than refuting it: the theory holds that apparent and functional coupling can dissociate and are not separable by observation, not that every exchange is $\mathcal{D}_2$ -like. If compliance is statistically indistinguishable between intact and ablated while responses remain fluent and on-topic, that is the dissociation, and it replaces the prescription in [Section 9](#) with a reported finding. Either outcome is reported as measured.